Oral Cancer Screening Report

AI-Assisted Mucosal Tissue Classification | Educational Prototype

23 October 2026 19:17

Version 1.0 | Swin-T Model

Classification Result

CANCER

Model Confidence: 99.7%
Logit Margin: N/A
Date / Time: 23 October 2026 19:17

1. About This Screening Tool

This report is produced by an AI-assisted oral mucosal tissue classifier based on a fine-tuned Swin Transformer (Swin-T) model trained on intraoral clinical photographs. The model distinguishes between oral mucosal tissue presentations that may or may not exhibit markers consistent with oral squamous cell carcinoma (OSCC). Classification follows the WHO 5th Edition taxonomy.

2. Image Quality Assessment

Metric	Measured Value	Acceptable Range	Status
Brightness (Luminance)	N/A / 255	55 – 225	N/A
Contrast (Std Dev)	N/A	≥ 16	N/A
Sharpness (Edge Var)	N/A	≥ 28	N/A
Logit Separation Margin	N/A	≥ 3.2 (reliable)	N/A

3. AI Model Explainability

How the classifier reached this result:

Architecture:
Swin Transformer (Swin-T) — a hierarchical vision transformer that splits the image into non-overlapping patches and learns local-to-global tissue features via shifted window attention.

Training data:
Fine-tuned on a labelled dataset of intraoral clinical photographs annotated for the presence or absence of malignant mucosal lesions.

Temperature scaling:
Raw logits are divided by T=1.5 before softmax, which prevents overconfident probability estimates that are common in small fine-tuned models.

Logit margin gate:
The absolute difference between the two class logits is N/A. A margin below 3.2 triggers an INCONCLUSIVE result to avoid false high-confidence predictions on non-intraoral images (e.g. face, clothing, teeth).

What the model looks for:
The model attends to colour irregularities, surface texture changes, mucosal pallor or erythroplakia patterns, and tissue boundary disruptions consistent with OSCC lesions.

Limitations:
This prototype cannot assess depth of invasion (DOI), perineural invasion, or lymph node status — all of which require histopathological examination. A positive result must be confirmed by a qualified oral oncologist.

4. Clinical Risk Interpretation Guide

AI Output	Interpretation	Recommended Action
NORMAL / BENIGN	No markers of malignant mucosal change detected	Routine dental follow-up; report any persistent lesions > 3 weeks to a clinician
CANCER / OSCC	Features consistent with	Immediate referral to oral oncology. Biopsy